Chapter 13

Algebraic fractions

nothing new: the same fraction moves, done twice side by side

numbers

$$\frac{2}{3}+\frac{4}{5}=\frac{2\cdot 5+3\cdot 4}{3\cdot 5}$$

letters

$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}$$

identical moves, the letters just sit where the numbers were

An **algebraic fraction** is a fraction with letters in it, like $\frac{3}{x}$ or $\frac{x+1}{x-2}$.

That's the only difference from Chapter 2, so the **rules are Chapter 2's rules**: the same three formulas for add/subtract, multiply and divide, and each section states its rule in a box before any example. Many moves are shown in **two columns**: numbers you already trust on the left, the same move with letters on the right.

One new idea does most of the work here: which parts of a fraction are **glued** together (multiplied) and which are only **loose** (added). You may cancel glued parts, and must never cancel loose ones.

Goes with **Lecture 1** (combining fractions) and **Lecture 2** (common mistakes, the 1/x = 2/y − 3/z example).

÷ Letters in a fraction, and the one number it hates

$\frac{x+1}{x-2}$ means **(the whole top) ÷ (the whole bottom)**. The fraction bar acts like a pair of brackets around the top and another around the bottom:

$$\frac{x+1}{x-2} = (x+1) \div (x-2)$$ ← *the bar groups each part*

To get a value, substitute (Chapter 8). With $x = 4$: $\frac{4+1}{4-2} = \frac{5}{2} = 2.5$.

The one number it hates. With $x = 2$ the bottom becomes $2 - 2 = 0$, and dividing by zero means nothing (Chapter 1: share 3 sweets among 0 people?). So this fraction only makes sense when $x \neq 2$. Whenever a letter sits on the bottom, ask: **which value makes the bottom zero?** That value is banned.

$$\frac{3}{x}$$
$x \neq 0$

$$\frac{5}{x+4}$$
$x \neq -4$

$$\frac{x}{(x-1)(x+3)}$$
$x \neq 1,\ x \neq -3$

÷ The three formulas, with letters

Exactly Chapter 2's formulas. Now a, b, c, d can be expressions like $x+1$; put each one in **brackets** when you plug it in.

Operation	Formula	Example with letters
Add / subtract	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$	$\frac{3}{x} - \frac{2}{y} = \frac{3y-2x}{xy}$
Multiply	$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$	$\frac{x+1}{x} \times \frac{x^2}{x+1} = \frac{(x+1)x^2}{x(x+1)} = x$
Divide	$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$	$\frac{x}{4} \div \frac{x^2}{6} = \frac{6x}{4x^2} = \frac{3}{2x}$

RULE: what's new with letters

Brackets: in $ad - bc$, b and c may be whole expressions; the minus applies to all of bc.

Banned values: any value that makes a bottom zero is not allowed.

Tidying the answer: factorise top and bottom, then cancel common **factors** (never terms): $\frac{x^2+3x}{x+3} = \frac{x(x+3)}{x+3} = x$.

÷ Cancelling: only glued parts come off

RULE: simplifying (cancelling)

Factorise the top and the bottom completely first.

Cancel a factor only if it **multiplies the whole top and the whole bottom**. Cancel factors, **never terms**: in $\frac{x+3}{3}$ the 3s do not cancel.

A banned value of the original fraction stays banned after cancelling.

Simplifying a fraction means dividing top and bottom by the **same thing**. Break both into atoms (as in Chapter 12) and cross out matching pairs:

with numbers

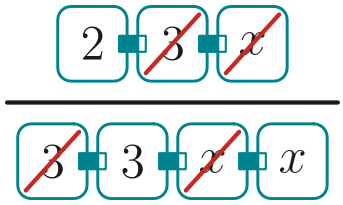
$$\frac{12}{18} = \frac{2 \cdot \cancel{6}}{3 \cdot \cancel{6}} = \frac{2}{3}$$

with letters

$$\frac{6x}{9x^2} = \frac{2 \cdot \cancel{3}\cancel{x}}{3x \cdot \cancel{3}\cancel{x}} = \frac{2}{3x}$$

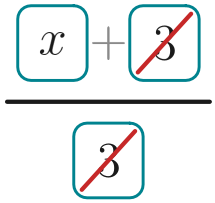
the same 6 (or 3x) on top and bottom

Here's the **glue** picture. The teal bars mean "multiplied", so those atoms are stuck to the whole top or the whole bottom. Dividing top and bottom by one of them is fine:



every atom is glued (x)
so a matching pair can be
lifted off: left with 2/(3x) ✓

Now a **loose** part. Can the 3s in $\frac{x+3}{3}$ cancel?



NO. The top 3 is only added on (loose).
It isn't multiplying the whole top.
Test x = 6: (6 + 3)/3 = 3, but "x" = 6 ✗

Fix: factorise first, to turn loose parts into glued ones.

with numbers

with letters

$$\frac{6+9}{3} = \frac{3(2+3)}{3} = 5$$

$$\frac{15}{20} = \frac{3 \cdot \cancel{5}}{4 \cdot \cancel{5}}$$

$$= \frac{3}{4}$$

$$\frac{x^2+3x}{x+3} = \frac{x\cancel{(x+3)}}{\cancel{(x+3)}} = x$$

factorise the top: now (x + 3) is glued on

$$\frac{x^2-9}{x^2+x-6} = \frac{(x-3)\cancel{(x+3)}}{(x-2)\cancel{(x+3)}}$$

top: squares; bottom: Δ = 25, pair 3, −2

$$= \frac{x-3}{x-2}$$

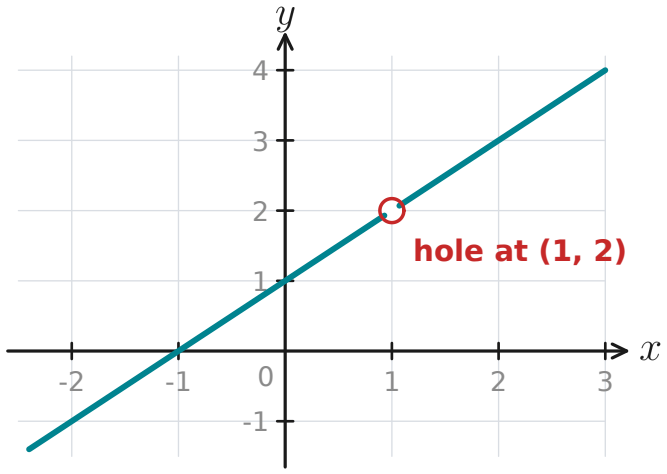
Rule of thumb: if there's a + or − showing on the top or bottom, factorise before you cancel anything.

Cancelling doesn't un-ban a number. Take $\frac{x^2-1}{x-1}$. The top is a difference of squares, so it cancels:

$$\frac{x^2-1}{x-1} = \frac{(x+1)\cancel{(x-1)}}{\cancel{(x-1)}} = x+1$$

✎ YOUR GO (pencil first)

The original fraction still has x − 1 on the bottom. Which x is banned? On the line y = x + 1, mark the one point the fraction can't give.



x	(x² − 1) ÷ (x − 1)
0.9	1.9
0.99	1.99
1	**0 ÷ 0: no value**
1.01	2.01
1.1	2.1

creeps up to 2 from both sides, but never actually lands on it

So the honest answer is $\frac{x^2-1}{x-1} = x+1$ **for** $x \neq 1$. The graph is the whole straight line with **one point punched out**, drawn as an open circle. Cancelling hides the $x-1$, but the ban stays with the original fraction.

÷ Back-to-front brackets: a − b = −(b − a)

RULE: reversing a subtraction

$a - b = -(b - a)$. Swapping the order of a subtraction only changes the sign: $5 - 3 = 2$ and $3 - 5 = -2$.

So $\frac{a-b}{b-a} = -1$ (as long as $a \neq b$).

When a top and bottom bracket are the same apart from order, take out -1 from one of them, then cancel.

with numbers

$$\frac{7-3}{3-7} = \frac{4}{-4} = -1$$

with letters

$$\frac{x-3}{3-x} = \frac{-(3-x)}{3-x} = -1$$

the top is −1 × the bottom

Example: $\frac{2-x}{x^2-4}$. The bottom factorises to $(x-2)(x+2)$, and $2 - x = -(x - 2)$:

$$\frac{2-x}{x^2-4} = \frac{-(x-2)}{(x-2)(x+2)}$$ ← *take out −1 on top; factorise the bottom*

$$= \frac{-1}{x+2} = -\frac{1}{x+2}$$ ← *cancel (x − 2); x ≠ 2, x ≠ −2*

Test $x = 0$: $\frac{2}{-4} = -0.5$ and $-\frac{1}{2} = -0.5$ ✓.

✗ Common mistakes

✗ $\frac{x+3}{3} = x$

3 is not a factor of the whole top: no cancelling

✓ $\frac{x+3}{3} = \frac{x}{3} + 1$

✗ $\frac{x^2+x}{x} = x^2 + 1$

factorise first: x is a factor of the WHOLE top

✓ $\frac{x(x+1)}{x} = x + 1$

✗ $\dfrac{x-3}{3-x}=1$

reversed brackets differ by a factor of −1

✓ $\dfrac{x-3}{3-x}=-1$

÷ Multiplying and dividing: the easy pair

RULE: multiplying and dividing algebraic fractions

Multiply: factorise every top and bottom, cancel any factor that appears on a top and a bottom (even in different fractions), then multiply tops together and bottoms together.

Divide: keep the first fraction, flip the second (its reciprocal), then multiply as above. **Flip before you cancel.**

Multiply: tops times tops, bottoms times bottoms. Divide: flip the second fraction (its reciprocal) and multiply. Exactly Chapter 2:

with numbers

$$\frac{2}{3}\times\frac{4}{5}=\frac{8}{15}$$

$$\frac{2}{3}\div\frac{4}{5}=\frac{2}{3}\times\frac{5}{4}$$

with letters

$$\frac{a}{b}\times\frac{c}{d}=\frac{ac}{bd}$$

tops × tops, bottoms × bottoms

$$\frac{a}{b}\div\frac{c}{d}=\frac{a}{b}\times\frac{d}{c}$$

flip the second, then multiply

A full example in tiny steps: $\frac{2x}{3}\times\frac{9}{4x^2}$

$$\frac{2x\cdot 9}{3\cdot 4x^2}$$ ← tops together, bottoms together

$$=\frac{18x}{12x^2}$$ ← 2 × 9 = 18; 3 × 4x² = 12x²

$$=\frac{3\cdot \cancel{6x}}{2x\cdot \cancel{6x}}=\frac{3}{2x}$$ ← cancel the glued 6x

Dividing: $\frac{x}{4}\div\frac{x^2}{6}$

$$= \frac{x}{4} \times \frac{6}{x^2} = \frac{6x}{4x^2}$$ ← flip the second fraction and multiply

$$= \frac{3}{2x}$$ ← cancel 2x from top and bottom

Test with $x = 3$: $\frac{3}{4} \div \frac{9}{6} = 0.75 \div 1.5 = 0.5$, and $\frac{3}{6} = 0.5$ ✓.

Dividing when you must factorise first: $\frac{x^2-9}{x+2} \div \frac{x-3}{x^2+2x}$

$$= \frac{x^2 - 9}{x + 2} \times \frac{x^2 + 2x}{x - 3}$$ ← flip the second, then ×

$$= \frac{(x-3)(x+3)}{x+2} \times \frac{x(x+2)}{x-3}$$ ← factorise every top and bottom

$$= \frac{\cancel{(x-3)}(x+3)}{\cancel{x+2}} \times \frac{\cancel{x(x+2)}}{\cancel{x-3}} = x(x+3)$$

cancel across the ×

Test $x = 1$: $\frac{-8}{3} \div \frac{-2}{3} = 4$ and $1 \times 4 = 4$ ✓.

÷ Adding and subtracting

RULE: adding and subtracting

$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$, with each top and bottom in brackets. This always works.

If the bottoms share a factor (like x and $2x$), the answer will contain that factor to cancel at the end. The optional procedure below avoids that by using a smaller bottom.

PROCEDURE: optional: the lowest common denominator (LCD)

1. **Factorise every bottom.**
2. The **LCD** is made of every different factor, each taken the **most times it appears in any one bottom**. (If the bottoms share no factor, the LCD is just the bottoms multiplied.)
3. Multiply the top and bottom of each fraction by whatever its bottom is **missing** from the LCD.
4. Add or subtract the tops (second top in brackets when subtracting). Keep the LCD.
5. Factorise the new top and cancel if you can.

with numbers

$$\frac{1}{3} + \frac{1}{4} = \frac{1 \cdot 4 + 3 \cdot 1}{3 \cdot 4}$$

$$= \frac{7}{12}$$

with letters

$$\frac{1}{x} + \frac{1}{y} = \frac{1 \cdot y + x \cdot 1}{xy}$$

ad + bc over bd

$$= \frac{y + x}{xy}$$

That's Lecture 1's formula. (Its subtraction slide prints a + where the − should be; the appendix lists it.) When the bottoms **share a factor** it gives a bigger bottom than needed and you cancel at the end; the LCD is the optional way to skip that.

Shared factor: $\frac{1}{x} + \frac{1}{2x}$. The bottoms are x and $2 \times x$. The factors are 2 (at most once) and x (at most once), so the LCD is $2x$, not $2x^2$:

$$\frac{1}{x}+\frac{1}{2x}=\frac{1\times 2}{x\times 2}+\frac{1}{2x}$$ ← x is missing a 2

$$=\frac{2+1}{2x}=\frac{3}{2x}$$ ← same bottoms: add the tops

(Multiplying the bottoms gives $\frac{2x+x}{2x^2}=\frac{3x}{2x^2}$, which still has to be cancelled down to $\frac{3}{2x}$.)

Factorise first: $\frac{1}{x+1}+\frac{2}{x^2-1}$. The second bottom is a difference of two squares, $(x-1)(x+1)$, and it already contains $x+1$:

$$\text{LCD}=(x-1)(x+1)$$ ← (x + 1) at most once, (x − 1) at most once

$$\frac{1(x-1)}{(x+1)(x-1)}+\frac{2}{(x-1)(x+1)}$$ ← the first is missing (x − 1)

$$=\frac{x-1+2}{(x-1)(x+1)}=\frac{x+1}{(x-1)(x+1)}$$ ← add the tops

$$=\frac{1}{x-1}$$ ← cancel (x + 1); x ≠ ±1

Test $x=2$: $\frac{1}{3}+\frac{2}{3}=1$ and $\frac{1}{1}=1$ ✓.

Bottoms with brackets: $\frac{2}{x+1}+\frac{3}{x-2}$. Treat each bottom as one glued lump.

$$\frac{2(x-2)}{(x+1)(x-2)}+\frac{3(x+1)}{(x+1)(x-2)}$$ ← each × the other bottom

$$=\frac{2x-4+3x+3}{(x+1)(x-2)}$$ ← add the tops; expand

$$=\frac{5x-1}{(x+1)(x-2)}$$ ← collect like terms

Test $x=0$: $\frac{2}{1}+\frac{3}{-2}=0.5$ and $\frac{-1}{(1)(-2)}=0.5$ ✓.

÷ The minus-sign trap

RULE: subtracting a fraction

The minus applies to the **whole top** of the fraction after it. Put that top in **brackets**, then multiply out: $-(x-1) = -x+1$.

When you subtract a fraction, the minus applies to its **whole top**. Keep the top in a bracket until the minus has been multiplied in:

$\frac{x+3}{2} - \frac{x-1}{3}$

$$= \frac{3(x+3) - 2(x-1)}{6}$$ ← common bottom 6; brackets kept!

$$= \frac{3x+9-2x+2}{6}$$ ← −2 × −1 = +2 (the trap is writing −2)

$$= \frac{x+11}{6}$$

Test $x = 1$: $\frac{4}{2} - \frac{0}{3} = 2$ and $\frac{12}{6} = 2$ ✓. Writing $-2x-2$ would give $\frac{x+7}{6}$, which is $\frac{8}{6}$ at $x = 1$ ✗.

÷ Whole numbers, and fractions inside fractions

RULE: a whole number (or expression) with a fraction

Write the whole number over the other fraction's bottom: $1 = \frac{x+1}{x+1}$, $2 = \frac{2x}{x}$. Then add or subtract as usual.

with numbers

$$1 - \frac{3}{4} = \frac{4}{4} - \frac{3}{4} = \frac{1}{4}$$

with letters

$$1 - \frac{3}{x+1} = \frac{x+1}{x+1} - \frac{3}{x+1}$$

1 written with the same bottom

$$= \frac{x+1-3}{x+1} = \frac{x-2}{x+1}$$

PROCEDURE: a fraction inside a fraction (a complex fraction)

1. Find the LCD of **all** the small fractions inside.
2. Multiply the **top and the bottom of the big fraction** by it. Every small fraction clears.
3. Simplify: factorise and cancel.

Example: $\frac{1+\frac{1}{x}}{1-\frac{1}{x}}$. The only small bottom is x, so multiply top and bottom by x:

$$\frac{\left(1+\frac{1}{x}\right)\times x}{\left(1-\frac{1}{x}\right)\times x} = \frac{x+1}{x-1}$$ ← *x × 1 = x and x × 1/x = 1*

Example: $\frac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{xy}}$. The LCD of x, y and xy is xy:

$$\frac{\left(\frac{1}{x}+\frac{1}{y}\right)\times xy}{\frac{1}{xy}\times xy} = \frac{y+x}{1} = x+y$$ ← *each small bottom cancels*

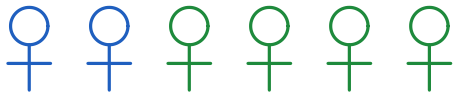
Test $x=2$, $y=3$: $\frac{1/2+1/3}{1/6} = \frac{5/6}{1/6} = 5$ and $2+3=5$ ✓.

÷ Splitting a fraction: the top may split, the bottom may not

Lecture 2's "common mistakes (2)" slide warns: $\frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$. Think of sweets:



12 sweets



2 friends arrive, then 4 more

$$\frac{12}{2+4} = \frac{12}{6} = 2 \text{ each} \qquad \frac{12}{2} + \frac{12}{4} = 6 + 3 = 9 \quad \times$$

Nine sweets each would need 54 sweets! Sharing among a **sum** of people is not the same as sharing among each group separately. But splitting the **top** is fine: 12 red + 6 blue sweets among 3 people is 4 red + 2 blue each.

allowed: split the top

$$\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$$

not allowed: split the bottom

$$\frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$$

The same slide's other warning is the root version: $\sqrt{a^2+b^2} \neq a+b$. Test $a=3$, $b=4$: $\sqrt{25}=5$, not 7.

÷ The lecture's letters-only example

Lecture 2 asks you to solve $\frac{1}{x} = \frac{2}{y} - \frac{3}{z}$ for z. The heart of it is one subtraction of fractions, which is this chapter's skill. On the way, you meet $\frac{2}{y} - \frac{1}{x}$:

with numbers

$$\frac{2}{5} - \frac{1}{3} = \frac{2 \cdot 3 - 5}{15}$$

with letters

$$\frac{2}{y} - \frac{1}{x} = \frac{2x - y}{xy}$$

a/b − c/d = (ad − bc)/bd

Chapter 14 finishes the job (flip both sides, multiply by 3) and gets $z = \frac{3xy}{2x-y}$. With $x=1$, $y=1$: $z=3$, and indeed $\frac{1}{1} = \frac{2}{1} - \frac{3}{3}$ ✓.

÷ Equations with algebraic fractions

An **equation** has an = sign, so you may multiply **both sides** by the same thing (Part 0's balance: do the same to both sides). That clears every fraction in one go. Chapters 14 and 15 do rearranging and solving in full; this is the fraction part.

PROCEDURE: solving an equation with fractions

1. Write down the **banned values** (anything that makes a bottom zero).
2. Multiply **every term on both sides** by the LCD of all the bottoms.
3. Solve what's left (no fractions now).
4. **Reject any answer that is a banned value**, then check the rest in the original equation.

Example: $\frac{3}{x} + \frac{1}{2} = 2$. Banned: $x = 0$. LCD: $2x$.

$$2x \cdot \frac{3}{x} + 2x \cdot \frac{1}{2} = 2x \cdot 2 \quad \leftarrow \times 2x, \text{ every term}$$

$$6 + x = 4x \;\Rightarrow\; 6 = 3x \;\Rightarrow\; x = 2 \quad \leftarrow \text{solve}$$

Check: $\frac{3}{2} + \frac{1}{2} = 2$ ✓, and 2 is not banned.

Example with a rejected answer: $\frac{x}{x-2} = \frac{2}{x-2} + 3$. Banned: $x = 2$.

$$x = 2 + 3(x - 2) \quad \leftarrow \times (x - 2), \text{ every term}$$

$$x = 3x - 4 \;\Rightarrow\; x = 2 \quad \leftarrow \text{solve}$$

$$x = 2 \text{ is banned: no solution} \quad \leftarrow \text{it makes the bottom 0}$$

Multiplying by $x - 2$ is multiplying by 0 when $x = 2$, which can create a fake answer. That's why step 4 is not optional.

✗ Common mistakes

✗ $\frac{3}{x} + \frac{1}{2} = 2 \;\rightarrow\; 6 + x = 2$ ✓ $6 + x = 4x$

multiply EVERY term by the LCD, including the 2 on the right

Algebraic fraction rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Banned values	a fraction is undefined where its bottom is 0; cancelling doesn't remove the ban
Simplify	$\dfrac{x^2+3x}{x+3} = \dfrac{x\cancel{(x+3)}}{\cancel{(x+3)}} = x$ factorise top and bottom first; cancel **factors**, never terms
Back-to-front brackets	$a-b=-(b-a), \qquad \dfrac{a-b}{b-a}=-1$
Add / subtract	$\dfrac{a}{b}\pm\dfrac{c}{d}=\dfrac{ad\pm bc}{bd}$ always works; optional LCD (factorise the bottoms, each factor the most times it appears) gives a smaller bottom
Subtracting	$\dfrac{x+3}{2}-\dfrac{x-1}{3}=\dfrac{3(x+3)-2(x-1)}{6}$ second top in brackets
Multiply	$\dfrac{a}{b}\times\dfrac{c}{d}=\dfrac{ac}{bd}$ multiply across; factorise and cancel first
Divide	$\dfrac{a}{b}\div\dfrac{c}{d}=\dfrac{a}{b}\times\dfrac{d}{c}=\dfrac{ad}{bc}$ flip the second only
Splitting	$\dfrac{a+b}{c}=\dfrac{a}{c}+\dfrac{b}{c}, \qquad \dfrac{a}{b+c}\neq\dfrac{a}{b}+\dfrac{a}{c}$ the top may split, the bottom may not
Whole number with a fraction	$1-\dfrac{3}{x+1}=\dfrac{x+1-3}{x+1}$
Fraction inside a fraction	multiply the big top and big bottom by the LCD of the small fractions
Equations with fractions	× every term by the LCD, solve, **reject banned values**, check

÷ Twin columns: your turn

Answers in grey. Check each by substituting a number such as x = 3.

When is it undefined?

1. $7/(x - 5)$
2. $x/(x^2 - 4)$

1. x = 5 2. x = 2 or x = −2

Cancel (factorise first!)

3. $10x^2/(15x)$
4. $(4x + 8)/4$
5. $(x^2 - 1)/(x + 1)$
6. $(x^2 + 5x + 6)/(x + 2)$
7. $(x + 5)/5$

3. 2x/3 4. x + 2 5. x − 1 6. x + 3 7. can't cancel: it's x/5 + 1

Multiply and divide

8. $(3x/4) \times (8/x^2)$
9. $(x/2) \div (x/10)$
10. $((x + 1)/x) \times (x^2/(x + 1))$

8. 6/x 9. 5 10. x

Add and subtract

11. $1/x + 2/x$
12. $1/x + 1/(2x)$
13. $3/x - 2/y$
14. $1/(x + 1) + 1/(x - 1)$
15. $(x + 1)/2 - (x - 3)/4$

11. 3/x 12. 3/(2x) (LCD 2x) 13. (3y − 2x)/(xy) 14. 2x/((x + 1)(x − 1)) 15. (x + 5)/4

Signs, nested fractions, equations

16. $(x - 5)/(5 - x)$
17. $1 - 2/(x + 3)$
18. $(1 + 2/x)/(1 - 4/x^2)$
19. $((x^2 - 4)/(x + 1)) \div ((x - 2)/(x^2 + x))$
20. Solve $4/x - 1/3 = 1$

16. −1 17. (x + 1)/(x + 3) 18. x/(x − 2) (× x², then cancel x + 2) 19. x(x + 2) 20. x = 3 (× 3x: 12 − x = 3x)

Find the unknown

21. $\frac{3}{(x-1)} + \frac{A}{(x+1)} = \frac{(5x+1)}{(x^2-1)}$. Find A.

22. $\frac{(x^2+kx+6)}{(x+2)}$ simplifies to $x + 3$. Find k.

23. $\frac{1}{x} + \frac{1}{(kx)} = \frac{5}{(4x)}$. Find k.

24. $\frac{(x+2)}{(x-3)}$: for which x is it undefined, and for which is it 0?

25. $\frac{a}{(x-2)} - \frac{1}{x} = \frac{(x+2)}{(x(x-2))}$. Find a.

21. A = 2 (3(x + 1) + A(x − 1) = 5x + 1) 22. k = 5 ((x + 2)(x + 3) = x² + 5x + 6) 23. k = 4 (× x: 1 + 1/k = 5/4) 24. undefined at x = 3 (bottom 0), zero at x = −2 (top 0) 25. a = 2 (ax − (x − 2) = x + 2)